Deterministic-weight recursive-tree path-length martingale limit

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Editorial note. Complete original mathematical paper64–1652, all native author proof paragraphs and embedded bibliography retained. Original source line1 pdfoutput is a PDFTeX-only engine flag and is outside rendering preamble2–41; no mathematical text omitted.

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Editorial note. Original1518–1521 AI-assisted revision disclosure retained literally: ChatGPT for language/literature comparison and checking/presenting arguments, authors responsible. AI_SCOPE.json records scope; no AI-free statement, undisclosed proof-origin inference or AI-dependency tracing.

Editorial note. One unified general-weight fluctuation theorem. Power-weight alpha regimes, explicit coefficients, exact generator and root-traversal applications remain complete context, not additional counts. Exact original CC BY4.0 retained.

Editorial note. Authors own mathematics. No proof repair, new assumptions, mathematical review, counterexample search, simulation/replay, content hashes, version comparisons or certification.

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Allowed standard prerequisites (editorial).

- Original weighted recursive-tree filtration and classical martingale convergence/Kronecker setup. All native positivity, deterministic-weight, p_n bound and probabilistic conventions retained literally with cited background.. Complete original model-specific innovation, summability and remainder proof supplied without replay or independent certification.

1 Introduction

Random recursive trees are standard probabilistic models of incrementally generated rooted structures. Vertices arrive one at a time, and every new vertex stores one link to an earlier vertex. The uniform model has a long history in combinatorial probability and the analysis of

algorithms; see Szymański (1990), Smythe and Mahmoud (1995), and Drmota (2009). The total path length

$$T_n := \sum_{i=1}^{n} D_i,$$

where D_i is the depth of vertex i, measures how far vertices lie from the root in aggregate. Its expectation is determined by individual depth means, whereas its fluctuations depend on the joint growth of the tree. We give a direct martingale-innovation analysis that applies without a regular cumulative-weight asymptotic and makes these fluctuations explicit.

Total path length also has a direct cost interpretation. Tracing parent pointers from vertex i to the root takes exactly D_i pointer traversals. Thus T_n is the cumulative cost of performing this operation once from every vertex, and T_n/n is its conditional average cost for a uniformly sampled stored vertex.

The classical uniform recursive tree already shows that global path length contains information not visible from the depth of one insertion. Mahmoud (1991) established an almost-sure and quadratic-mean limit theorem for its centered internal path length, and Dobrow and Fill (1999) developed the limiting distribution in substantially greater detail. Total path length also plays a central role in the average-case analysis of binary search trees and Quicksort; see Knuth (1998) and Flajolet and Sedgewick (2009). In a random binary search tree the expected internal path length equals

$$2(n+1)H_n - 4n,$$

while its variance is asymptotic to $(7 - 2\pi^2/3)n^2$.

The present paper concerns nonuniform recursive growth. In a weighted recursive tree, vertex i receives a deterministic positive weight w_i and a later vertex chooses its parent with probability proportional to these weights. General weighted recursive trees have been studied from several viewpoints; see, for example, Borovkov and Vatutin (2006), Hiesmayr and Işlak (2020), and Sénizergues (2021). We focus on the power-weight family

$$w_i = i^{\alpha}, \qquad \alpha \in \mathbb{R}.$$

This model interpolates between preference for recent vertices ($\alpha > 0$), uniform attachment ($\alpha = 0$), and preference for old vertices ($\alpha < 0$).

The linear rule $\alpha = 1$ appears as a label-attraction model in Gałązka et al. (2020), where degree statistics and degree counts were studied, and as the young-age preferential-attachment tree of Lyon and Mahmoud (2020), where insertion depth, outdegrees, and leaves were analyzed. Lyon and Mahmoud (2022) subsequently treated insertion depth in the full power-weight family. The Bernoulli representation underlying insertion depth is implicit in the aggregation-tree coupling of Curien and Haas (2017, equation (2.2)) and is explicit for arbitrary positive deterministic weights in Mailler and Uribe Bravo (2019, Corollary 8); see also the weighted-recursive-tree and general-affinity treatments in Sénizergues (2021); Nakata and Mahmoud (2024). In particular, for $\alpha > -1$ the typical insertion-depth scale is logarithmic, at $\alpha = -1$ it is of order $\log\log n$, and for $\alpha < -1$ it remains on a finite scale. In the special young-age case $\alpha = 1$, the insertion depth has exactly the same distribution as the insertion depth in a random binary search tree (Lyon and Mahmoud, 2020).

Path-length functionals have also been studied in several other nonuniform or weighted tree models. Examples include weighted path lengths in increasing trees (Kuba and Panholzer, 2007), internal path length in Hoppe trees (Leckey and Neininger, 2013), and path length in weighted random b-ary recursive trees (Munsonius and Rüschendorf, 2011). These uses of "weighted" refer to different tree families or to weights on path functionals; they do not give the total path-length theory for deterministic label weights $w_i = i^{\alpha}$ considered here.

Relation to the weighted-recursive-tree profile theory. For $\alpha > -1$, the weights $w_i = i^\alpha$ satisfy the assumptions of the profile analysis of Sénizergues (2021), with $\gamma = \alpha + 1$. In the section "Height and profile of WRT", the normalized profile transform

$$N_n(z) = n^{-1-\gamma(e^z-1)} \sum_{i=1}^{n} e^{zD_i}$$

is shown to converge almost surely, locally uniformly on a complex neighbourhood of zero, to a random analytic function. Since $N_n'(0) = T_n/n - \gamma \log n$, differentiation gives an almost-sure limit for this centered average depth. The convergence of $(T_n - \mathbb{E}T_n)/n$ in this regime follows after accounting for the deterministic constant in its mean asymptotics. The weighted-depth martingale used below also has a direct antecedent in the section "Study of the Laplace transform of the weighted profile". With normalization starting at index 2, differentiating

$$\mathcal{M}_n(z) = \frac{W_n^{-1} \sum_{i=1}^{n} w_i e^{zD_i}}{\prod_{i=2}^{n} (1 + (e^z - 1)p_i)}$$

at zero, with $A_n = \sum_{i=1}^{n} w_i D_i$, yields $\mathcal{M}_n'(0) = (A_n - \mathbb{E}A_n)/W_n$.

Contributions and scope. The model, the one-vertex Bernoulli representation, and the associated depth scales are known. The profile theory described above also implies the almost-sure centered path-length limit for power weights with $\alpha > -1$. We do not claim that limit or the underlying weighted-profile martingale as new. The results below provide a common elementary proof without prescribing a cumulative-weight asymptotic, together with explicit second-order calculations. More precisely:

(i) For arbitrary positive deterministic weights, we derive the exact identity

$$T_n - \mathbb{E}T_n = \sum_{j=2}^{n} (1 + (n-j)p_j)\xi_j$$

and an exact recurrence for $v_j = \mathbb{E}\xi_j^2$. These formulas describe how each attachment innovation contributes to total path length and its variance.

(ii) If $p_n = O(n^{-1})$, then $(T_n - \mathbb{E}T_n)/n$ converges almost surely and in L^2 to a nondegenerate limit, with

$$\mathrm{Var}(T_n) \sim n^2 \sum_{j=2}^{\infty} p_j^2 v_j.$$

The hypothesis covers all power weights, including $\alpha = -1$ and $\alpha < -1$, and also bounded oscillating weights for which W_n/n has no limit.

(iii) For $\alpha < -1$, the average depth has the explicit random limit

$$\frac{T_n}{n} \longrightarrow 1 + \frac{\sum_{i \geq 1} i^\alpha D_i}{\zeta(-\alpha)} \qquad \text{almost surely and in } L^2.$$

Thus the bounded expected average depth in this regime does not imply convergence to a deterministic constant.

(iv) For $\alpha = 1$, we solve the variance recurrence and obtain

$$\mathrm{Var}(T_n) \sim \left(8 - \frac{2\pi^2}{3}\right) n^2.$$

The coefficient is larger by exactly one than that of a random binary search tree, although the insertion-depth marginals and expected total path length coincide.

For completeness, we also obtain the exact expectation and its three power-weight regimes by summing the known one-vertex means, recover the classical uniform-tree benchmark, and give a simple exact parent sampler for linear weights. These supporting results place the fluctuation analysis in context.

Section 2 introduces the model and recalls the depth representation. Sections 3 and 4 give the linear-weight sampler and the power-weight expectation asymptotics. Sections 5 and 6 establish the exact innovation identities and the general limit theorem. Sections 7 and 8 identify the summable-regime limit and evaluate the linear-weight variance coefficient.

2 Weighted recursive trees and power weights

2.1 Definition

Let $w_1, w_2, \ldots > 0$ be deterministic weights and put

$$W_n := \sum_{i=1}^{n} w_i, \qquad p_j := \frac{w_j}{W_j}.$$

The weighted recursive tree is constructed on $\{1, \ldots, n\}$ from root 1. For $n \geq 2$, vertex n independently chooses a parent $K_n \in \{1, \ldots, n-1\}$ with

$$\mathbb{P}(K_n = i) = \frac{w_i}{W_{n-1}}, \qquad 1 \leq i < n. \tag{1}$$

All trees are coupled using the same infinite sequence of independent parent choices. Let $D_1 = 0$ and let D_n be the depth of vertex n. Then

$$D_n = 1 + D_{K_n}, \qquad n \geq 2. \tag{2}$$

The power-weight specialization is $w_i = i^\alpha$, $\alpha \in \mathbb{R}$, and is denoted by $T_n^{(\alpha)}$ when the tree itself is needed. For these weights,

$$p_j = \frac{j^\alpha}{\sum_{i=1}^{j} i^\alpha}. \tag{3}$$

In either setting $p_1 = 1$, and total path length is

$$T_n := \sum_{i=1}^{n} D_i. \tag{4}$$

2.2 Known insertion-depth representation

The following representation is a direct consequence of the ancestral-indicator representation in Mailler and Uribe Bravo (2019, Corollary 8), which holds for arbitrary positive weight sequences. Indeed, the parent K_n is sampled proportionally to the weights in the tree on $n-1$ vertices, and $D_n = D_{K_n} + 1$. An implicit precursor appears in the aggregation-tree coupling of Curien and Haas (2017, equation (2.2)). Later treatments include Sénizergues (2021); Lyon and Mahmoud (2022); Nakata and Mahmoud (2024). We include a short derivation for notation and self-containedness because the parameters p_j drive the entire path-length analysis; no novelty is claimed for this representation.

Proposition 2.1 (Bernoulli representation)**.** *For every $n \geq 2$,*

$$D_n \stackrel{d}{=} \sum_{j=1}^{n-1} B_j,$$

where $B_1, B_2, \ldots$ are independent Bernoulli variables satisfying $\mathbb{P}(B_j = 1) = p_j$. Equivalently,

$$\mathbb{E}z^{D_n} = \prod_{j=1}^{n-1}(1 - p_j + p_j z). \tag{5}$$

Consequently,

$$\mathbb{E}D_n = \sum_{j=1}^{n-1} p_j, \tag{6}$$

$$\mathrm{Var}(D_n) = \sum_{j=1}^{n-1} p_j(1 - p_j). \tag{7}$$

Proof. Let $F_n(z) := \mathbb{E}z^{D_n}$ and define

$$S_m(z) := \sum_{i=1}^{m} w_i F_i(z).$$

Conditioning on K_n in (2) gives

$$F_n(z) = z\frac{S_{n-1}(z)}{W_{n-1}}.$$

Therefore

$$S_n(z) = S_{n-1}(z) + w_n F_n(z) = S_{n-1}(z)\left(1 + \frac{zw_n}{W_{n-1}}\right),$$

and hence

$$\frac{F_{n+1}(z)}{F_n(z)} = \frac{W_{n-1} + zw_n}{W_n} = 1 + (z-1)\frac{w_n}{W_n} = 1 - p_n + p_n z.$$

Since $F_2(z) = z$ and $p_1 = 1$, induction yields (5). The moment formulas follow immediately. □

Remark 2.2 (Novelty boundary for depth)**.** *Proposition 2.1 and the resulting one-vertex depth moments are background. The analysis below concerns the accumulated global functional T_n, its martingale representation, and its fluctuations across insertion times.*

Lemma 2.3 (Asymptotics of the Bernoulli parameters)**.** *For power weights $w_n = n^\alpha$, the sequence (p_n) has the following behavior.*

(i) If $\alpha > -1$ and $c_\alpha := \alpha + 1$, then for some $\delta_\alpha > 0$,

$$p_n = \frac{c_\alpha}{n} + O(n^{-1-\delta_\alpha}), \tag{8}$$

and in particular

$$\sum_{n\geq 1}\left|p_n - \frac{c_\alpha}{n}\right| < \infty.$$

(ii) If $\alpha = -1$, then

$$p_n = \frac{1}{nH_n}, \tag{9}$$

where $H_n = \sum_{k=1}^{n} 1/k$.

(iii) If $\alpha < -1$, then

$$p_n \sim \frac{n^\alpha}{\zeta(-\alpha)} \tag{10}$$

and $\sum_{n\geq 1} p_n < \infty$.

Proof. For $\alpha > -1$, comparison of the power sum with its integral gives

$$W_n = \frac{n^{\alpha+1}}{\alpha+1}\begin{cases}1 + O(n^{-1}), & \alpha \geq 0,\\ 1 + O(n^{-(\alpha+1)}), & -1 < \alpha < 0.\end{cases}$$

Substitution in (3) gives (8), with $\delta_\alpha = \min\{1, \alpha+1\}$. The error is summable.

For $\alpha = -1$, $W_n = H_n$, which gives (9). If $\alpha < -1$, then $W_n \to \sum_{k\geq 1} k^\alpha = \zeta(-\alpha)$, proving (10); summability follows because $\alpha < -1$. □

3 An exact linear-time generator for the young-age tree

The power $\alpha = 1$ gives

$$W_{n-1} = \frac{n(n-1)}{2}$$

and therefore

$$\mathbb{P}(K_n = i) = \frac{2i}{n(n-1)}, \qquad 1 \leq i < n. \tag{11}$$

This is the young-age preferential tree of Lyon and Mahmoud (2020). The special linear weight admits a particularly simple exact generator.

Proposition 3.1 (Two-sample parent rule)**.** *Fix $n \geq 2$. Choose an ordered pair (U, V) uniformly from*

$$\{(u, v) : 1 \leq u, v \leq n,\ u \neq v\},$$

and set

$$K_n := \max\{U, V\} - 1. \tag{12}$$

Then K_n has the distribution (11)*.*

Proof. For $1 \leq i < n$, the event $K_n = i$ is the event $\max\{U, V\} = i + 1$. Exactly $2i$ ordered distinct pairs have maximum $i + 1$: either $U = i + 1$ and $V \in \{1, \ldots, i\}$, or conversely. Since there are $n(n-1)$ ordered distinct pairs in total,

$$\mathbb{P}(K_n = i) = \frac{2i}{n(n-1)}.$$

□

Implementation. An ordered distinct pair can be sampled without rejection. Draw $U \sim \mathrm{Unif}\{1, \ldots, n\}$ and $R \sim \mathrm{Unif}\{1, \ldots, n-1\}$ independently, and set

$$V = \begin{cases}R, & R < U,\\ R + 1, & R \geq U.\end{cases}$$

Then apply (12).

Corollary 3.2 (Generation complexity)**.** *In a unit-cost word-RAM model with $O(\log n)$-bit words and constant-time uniform integer sampling, the parent array of a young-age preferential tree of size n can be generated exactly in $\Theta(n)$ time and $\Theta(n)$ space. If the edges are streamed instead of stored, the extra working space is $O(1)$.*

The probabilistic results below do not rely on this sampler. Its role is to show that the distinguished young-age case also admits a direct randomized generation algorithm with constant work per insertion.

4 Expected total path length and the first-order phase transition

Throughout this section, $w_i = i^\alpha$ with fixed $\alpha \in \mathbb{R}$.

Let U_n be uniform on $\{1, \dots, n\}$, conditionally independent given the tree. If the operation Trace-To-Root repeatedly replaces a vertex by its parent until the root is reached, then its number of pointer traversals is D_{U_n}. Conditional on the tree,

$$\mathbb{E}(D_{U_n} \mid T_n^{(\alpha)}) = \frac{T_n}{n}. \tag{13}$$

Thus T_n/n is the quenched average query cost.

Theorem 4.1 (Exact expectation and phase transition)**.** *For every $\alpha \in \mathbb{R}$ and $n \geq 1$,*

$$\mathbb{E}T_n = \sum_{j=1}^{n-1} (n-j) p_j. \tag{14}$$

Moreover:

(i) If $\alpha > -1$ and $c_\alpha = \alpha + 1$, then

$$\frac{\mathbb{E}T_n}{n} = c_\alpha \log n + \kappa_\alpha + o(1), \tag{15}$$

where

$$\kappa_\alpha = c_\alpha \gamma - c_\alpha + \sum_{j=1}^{\infty} \left(p_j - \frac{c_\alpha}{j} \right). \tag{16}$$

(ii) If $\alpha = -1$, then there exists a finite constant κ_{-1} such that

$$\frac{\mathbb{E}T_n}{n} = \log\log n + \kappa_{-1} + o(1). \tag{17}$$

(iii) If $\alpha < -1$, then

$$\frac{\mathbb{E}T_n}{n} \longrightarrow \mu_\alpha, \qquad \mu_\alpha := \sum_{j=1}^{\infty} p_j < \infty. \tag{18}$$

Proof. By Proposition 2.1,

$$\mathbb{E}T_n = \sum_{i=1}^{n} \mathbb{E}D_i = \sum_{i=2}^{n} \sum_{j=1}^{i-1} p_j.$$

Interchanging the finite sums gives (14). Dividing by n,

$$\frac{\mathbb{E}T_n}{n} = \sum_{j=1}^{n-1} p_j - \frac{1}{n} \sum_{j=1}^{n-1} j p_j. \tag{19}$$

Suppose first that $\alpha > -1$. By Lemma 2.3,

$$\sum_{j=1}^{n-1} p_j = c_\alpha H_{n-1} + \sum_{j=1}^{\infty} \left(p_j - \frac{c_\alpha}{j} \right) + o(1),$$

while $j p_j \to c_\alpha$, so Cesàro's theorem gives

$$\frac{1}{n} \sum_{j=1}^{n-1} j p_j \to c_\alpha.$$

Using $H_{n-1} = \log n + \gamma + o(1)$ proves (15)–(16).

For $\alpha = -1$, $p_j = 1/(jH_j)$. Since

$$H_j = \log j + \gamma + O(j^{-1}),$$

we have, for $j \geq 2$,

$$\frac{1}{jH_j} = \frac{1}{j \log j} + O\Big(\frac{1}{j(\log j)^2}\Big).$$

The error is summable and the standard integral comparison gives

$$\sum_{j=2}^{n-1} \frac{1}{j \log j} = \log\log n + C + o(1)$$

for a finite constant C. On the other hand, $jp_j = 1/H_j \to 0$, so the second term of (19) tends to zero. This proves (17).

Finally, if $\alpha < -1$, then $\sum_j p_j = \mu_\alpha < \infty$ and $jp_j \to 0$ by Lemma 2.3. Hence the first term in (19) tends to μ_α, while the second tends to zero by Cesàro's theorem. □

The theorem can be read directly as an average-case complexity statement.

Corollary 4.2 (Average root-traversal complexity)**.** *The expected conditional average number of parent-pointer traversals satisfies*

$$\mathbb{E}\Big[\mathbb{E}(D_{U_n} \mid T_n^{(\alpha)})\Big] \sim \begin{cases} (\alpha+1)\log n, & \alpha > -1, \\ \log\log n, & \alpha = -1, \\ \mu_\alpha, & \alpha < -1. \end{cases}$$

Thus $\alpha = -1$ separates logarithmic and constant-order average traversal cost, with an iterated logarithm exactly at criticality.

4.1 Second-order expectation inside the old-age phase

The leading term for every $\alpha < -1$ is linear, but its deterministic correction has another threshold at $\alpha = -2$.

Proposition 4.3. *Let $\alpha < -1$.*

(i) If $-2 < \alpha < -1$, then

$$\mathbb{E}T_n = \mu_\alpha n - \frac{n^{\alpha+2}}{\zeta(-\alpha)(-\alpha-1)(\alpha+2)}(1 + o(1)).$$

(ii) If $\alpha = -2$, then

$$\mathbb{E}T_n = \mu_{-2} n - \frac{\log n}{\zeta(2)} + O(1).$$

(iii) If $\alpha < -2$, then

$$\mathbb{E}T_n = \mu_\alpha n - \lambda_\alpha + o(1), \qquad \lambda_\alpha := \sum_{j \geq 1} jp_j < \infty.$$

Proof. From (14),

$$\mu_\alpha n - \mathbb{E}T_n = n \sum_{j \geq n} p_j + \sum_{j=1}^{n-1} jp_j. \tag{20}$$

Write $c_\alpha := 1/\zeta(-\alpha)$. By Lemma 2.3,

$$p_j \sim c_\alpha j^\alpha.$$

If $-2 < \alpha < -1$, the standard power-sum and regularly varying tail estimates give

$$n \sum_{j \geq n} p_j \sim \frac{c_\alpha}{-\alpha - 1} n^{\alpha+2}, \qquad \sum_{j=1}^{n-1} j p_j \sim \frac{c_\alpha}{\alpha + 2} n^{\alpha+2}.$$

Substitution in (20) yields

$$\mu_\alpha n - \mathbb{E}T_n \sim \frac{n^{\alpha+2}}{\zeta(-\alpha)} \left(\frac{1}{-\alpha - 1} + \frac{1}{\alpha + 2} \right) = \frac{n^{\alpha+2}}{\zeta(-\alpha)(-\alpha - 1)(\alpha + 2)}.$$

At the boundary $\alpha = -2$ we need one term more than the equivalence $p_j \sim 1/(\zeta(2)j^2)$. The tail of the Basel series satisfies

$$W_j = \sum_{k=1}^{j} \frac{1}{k^2} = \zeta(2) - \frac{1}{j} + O(j^{-2}),$$

and therefore

$$\frac{1}{W_j} = \frac{1}{\zeta(2)} + \frac{1}{\zeta(2)^2 j} + O(j^{-2}).$$

Consequently,

$$p_j = \frac{1}{\zeta(2) j^2} + O(j^{-3}).$$

It follows that

$$n \sum_{j \geq n} p_j = O(1), \qquad \sum_{j=1}^{n-1} j p_j = \frac{1}{\zeta(2)} \log n + O(1),$$

which proves part (ii).

Finally, if $\alpha < -2$, then $\sum_{j \geq 1} j p_j < \infty$. Moreover,

$$n \sum_{j \geq n} p_j = O(n^{\alpha+2}) = o(1),$$

while

$$\sum_{j=1}^{n-1} j p_j \longrightarrow \lambda_\alpha.$$

Equation (20) now gives part (iii). □

4.2 The young-age mean and a first BST coincidence

For $\alpha = 1$,

$$p_j = \frac{2}{j + 1}.$$

Equation (14) then gives an exact closed form.

Corollary 4.4. *For the young-age preferential tree,*

$$\mathbb{E}T_n = 2(n + 1)H_n - 4n. \tag{21}$$

Proof. Substituting $p_j = 2/(j + 1)$ in (14) and summing gives

$$2 \sum_{j=1}^{n-1} \frac{n - j}{j + 1} = 2(n + 1)H_n - 4n.$$

□

The right-hand side of (21) is also the expected internal path length of a random binary search tree on n keys (Knuth, 1998; Flajolet and Sedgewick, 2009). Section 8 will show that the agreement stops at the first moment.

5 Weighted-depth martingales and exact fluctuation identities

In this section the weights $w_i > 0$ are arbitrary and deterministic; no growth assumption is imposed. Let $\mathcal{F}_n$ be the sigma-field generated by the parent choices $K_2, \dots, K_n$. Define the weighted depth sum

$$A_n := \sum_{i=1}^{n} w_i D_i. \tag{22}$$

By (2),

$$\mathbb{E}(D_{n+1} \mid \mathcal{F}_n) = 1 + \frac{A_n}{W_n}. \tag{23}$$

Introduce the innovation

$$\xi_{n+1} := D_{n+1} - \mathbb{E}(D_{n+1} \mid \mathcal{F}_n), \qquad n \geq 1. \tag{24}$$

Then

$$\mathbb{E}(\xi_{n+1} \mid \mathcal{F}_n) = 0.$$

We put $v_n := \mathbb{E}\xi_n^2$ for $n \geq 2$.

Proposition 5.1 (Weighted-depth martingale)**.** *Let*

$$\widetilde{A}_n := A_n - \mathbb{E}A_n.$$

Then

$$M_n := \frac{\widetilde{A}_n}{W_n} = \sum_{j=2}^{n} p_j \xi_j \tag{25}$$

is a square-integrable martingale.

Proof. Using $A_{n+1} = A_n + w_{n+1} D_{n+1}$ and (23)–(24), centering gives

$$\widetilde{A}_{n+1} = \left(1 + \frac{w_{n+1}}{W_n}\right) \widetilde{A}_n + w_{n+1} \xi_{n+1}.$$

Since $W_{n+1} = W_n + w_{n+1}$,

$$\frac{\widetilde{A}_{n+1}}{W_{n+1}} = \frac{\widetilde{A}_n}{W_n} + \frac{w_{n+1}}{W_{n+1}} \xi_{n+1} = M_n + p_{n+1} \xi_{n+1}.$$

As $M_1 = 0$, iteration proves (25). □

The martingale in Proposition 5.1 is the derivative at zero of the normalized weighted-profile martingale described in the introduction. We include its short direct proof to establish the notation needed for the exact innovation identities.

The same martingale gives an exact representation for total path length.

Theorem 5.2 (Exact innovation representation)**.** *For every $n \geq 2$,*

$$T_n - \mathbb{E}T_n = \sum_{j=2}^{n} \big(1 + (n-j)p_j\big) \xi_j. \tag{26}$$

Consequently,

$$\operatorname{Var}(T_n) = \sum_{j=2}^{n} \big(1 + (n-j)p_j\big)^2 v_j. \tag{27}$$

Proof. From $T_{n+1} = T_n + D_{n+1}$ and (23),

$$(T_{n+1} - \mathbb{E}T_{n+1}) - (T_n - \mathbb{E}T_n) = M_n + \xi_{n+1}.$$

Summing from 1 to $n-1$ and inserting $M_m = \sum_{j=2}^{m} p_j \xi_j$ gives

$$T_n - \mathbb{E}T_n = \sum_{j=2}^{n} \xi_j + \sum_{m=1}^{n-1} \sum_{j=2}^{m} p_j \xi_j = \sum_{j=2}^{n} (1 + (n-j)p_j)\xi_j.$$

Martingale differences are orthogonal in L^2, so (27) follows. □

The innovation variances have a simple universal recurrence.

Proposition 5.3 (Innovation variance recurrence)**.** *Let*

$$V_n := \mathrm{Var}(D_n) = \sum_{j=1}^{n-1} p_j(1-p_j).$$

Then

$$v_n = V_n - \sum_{j=2}^{n-1} p_j^2 v_j \tag{28}$$

and, for $n \geq 2$,

$$v_{n+1} = (1-p_n^2)v_n + p_n(1-p_n), \qquad v_2 = 0. \tag{29}$$

Proof. By the law of total variance,

$$\mathrm{Var}(D_n) = \mathbb{E}[\mathrm{Var}(D_n \mid \mathcal{F}_{n-1})] + \mathrm{Var}(\mathbb{E}(D_n \mid \mathcal{F}_{n-1})).$$

The first term is v_n. By (23) and Proposition 5.1, the second is

$$\mathrm{Var}(M_{n-1}) = \sum_{j=2}^{n-1} p_j^2 v_j,$$

which proves (28). Subtracting consecutive identities and using $V_{n+1} - V_n = p_n(1-p_n)$ gives (29). □

6 A linear-scale limit theorem for deterministic weights

We first state the limit theorem for a class of positive deterministic attachment weights. Let $w_1, w_2, \ldots > 0$, write $W_n = \sum_{i=1}^{n} w_i$ and $p_n = w_n/W_n$, and assume that

$$p_n = O(n^{-1}). \tag{30}$$

All random variables are defined on the coupled sequence of trees obtained from independent parent choices. We use the innovations ξ_n, their variances $v_n = \mathbb{E}\xi_n^2$, and the martingale $M_n = \sum_{j=2}^{n} p_j \xi_j$ from the preceding section.

Lemma 6.1 (Square summability for deterministic weights)**.** *Under assumption* (30)*,*

$$v_n = O(\log n), \qquad \sum_{j=2}^{\infty} p_j^2 v_j < \infty, \qquad \sum_{j=2}^{\infty} \frac{v_j}{j^2} < \infty. \tag{31}$$

Proof. The innovation-variance recurrence gives $v_{n+1} \le v_n + p_n$, with $v_2 = 0$. Hence

$$v_n \le \sum_{j=2}^{n-1} p_j = O(\log n).$$

Since $p_j = O(j^{-1})$, both series in (31) are bounded by a constant multiple of $\sum_{j=2}^{\infty}(\log j)/j^2$, apart from finitely many terms. □

Theorem 6.2 (Path-length limit for deterministic weights)**.** *Let $w_1, w_2, \ldots > 0$ be deterministic attachment weights satisfying* (30)*. Then the series*

$$Z := \sum_{j=2}^{\infty} p_j \xi_j \tag{32}$$

converges almost surely and in L^2, and

$$\frac{T_n - \mathbb{E}T_n}{n} \longrightarrow Z \qquad \text{almost surely and in } L^2. \tag{33}$$

Moreover, $\mathbb{E}Z = 0$, the limit is nondegenerate, and

$$\mathrm{Var}(T_n) \sim \sigma^2 n^2, \qquad \sigma^2 := \mathbb{E}Z^2 = \sum_{j=2}^{\infty} p_j^2 v_j \in (0, \infty). \tag{34}$$

Proof. By Lemma 6.1 and orthogonality of martingale differences,

$$\sup_n \mathbb{E}M_n^2 = \sum_{j=2}^{\infty} p_j^2 v_j < \infty.$$

The martingale convergence theorem therefore gives $M_n \to Z$ almost surely and in L^2, with $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = \sum_{j=2}^{\infty} p_j^2 v_j$.

The exact innovation representation in the preceding section yields

$$\frac{T_n - \mathbb{E}T_n}{n} = M_n + \frac{1}{n}\sum_{j=2}^{n} a_j \xi_j, \qquad a_j := 1 - j p_j. \tag{35}$$

Assumption (30) implies $\sup_{j \ge 2} |a_j| < \infty$. Consequently,

$$\sum_{j=2}^{\infty} \mathbb{E}\left(\frac{a_j \xi_j}{j}\right)^2 < \infty.$$

The martingale series $\sum_{j=2}^{\infty} a_j \xi_j / j$ thus converges almost surely. Kronecker's lemma gives

$$\frac{1}{n}\sum_{j=2}^{n} a_j \xi_j \longrightarrow 0 \qquad \text{almost surely.}$$

For convergence in L^2, orthogonality and Lemma 6.1 give

$$\mathbb{E}\left(\frac{1}{n}\sum_{j=2}^{n} a_j \xi_j\right)^2 = \frac{1}{n^2}\sum_{j=2}^{n} a_j^2 v_j = O\left(\frac{\log n}{n}\right) \longrightarrow 0.$$

Together with $M_n \to Z$, these estimates prove (33).

Since all weights are strictly positive, $0 < p_2 = w_2/(w_1 + w_2) < 1$ and $p_3 > 0$. The recurrence with $v_2 = 0$ gives $v_3 = p_2(1 - p_2) > 0$. Therefore

$$\sigma^2 \ge p_3^2 p_2 (1 - p_2) > 0,$$

which proves nondegeneracy. Finally, convergence in L^2 in (33), with both sides centered, implies $\mathrm{Var}(T_n)/n^2 \to \mathbb{E}Z^2 = \sigma^2$ and hence (34). □

Corollary 6.3 (Power-weight path-length limit)**.** *For every fixed $\alpha \in \mathbb{R}$, let $w_j = j^\alpha$ and $p_j = j^\alpha / W_j$. Then*

$$Z_\alpha := \sum_{j=2}^{\infty} p_j \xi_j, \qquad \frac{T_n - \mathbb{E}T_n}{n} \longrightarrow Z_\alpha \quad \text{almost surely and in } L^2, \tag{36}$$

where the series converges almost surely and in L^2, and

$$\mathrm{Var}(T_n) \sim \sigma_\alpha^2 n^2, \qquad \sigma_\alpha^2 := \mathbb{E}Z_\alpha^2 = \sum_{j=2}^{\infty} p_j^2 v_j \in (0, \infty). \tag{37}$$

Proof. For $\alpha > -1$, the power-sum estimate gives $p_j \sim (\alpha+1)/j$. At $\alpha = -1$, $p_j = 1/(jH_j) \le 1/j$. For $\alpha < -1$, the bound $W_j \ge w_1 = 1$ gives $p_j \le j^\alpha \le 1/j$. Thus (30) holds in every case, and the result follows from Theorem 6.2. □

Remark 6.4 (No regular cumulative-weight asymptotic is needed)**.** *For $w_n = 1 + \frac{1}{2}\sin(\log n)$ we have $\frac{1}{2} \le w_n \le \frac{3}{2}$, hence $p_n \le 3/n$. The sum–integral error is $O(\log n)$ because the absolute derivative of $\sin(\log x)$ is at most $1/x$. Integration therefore gives*

$$\frac{W_n}{n} = 1 + \frac{1}{4}\big(\sin(\log n) - \cos(\log n)\big) + O\Big(\frac{\log n}{n}\Big).$$

Thus W_n/n has no limit, and $W_n \sim cn^\gamma$ fails for every $c, \gamma > 0$. Theorem 6.2 nevertheless applies. Thus no polynomial cumulative-weight asymptotic is needed.

For the remainder of the paper we specialize again to $w_i = i^\alpha$.

6.1 Recovery of the classical uniform recursive tree

The case $\alpha = 0$ is the classical uniform recursive tree. It provides a useful exact benchmark for the general theory.

Corollary 6.5 (Uniform recursive tree benchmark)**.** *Let $\alpha = 0$. Then $p_j = 1/j$ and, for every $n \ge 2$,*

$$\mathbb{E}T_n = n(H_n - 1). \tag{38}$$

Moreover,

$$\frac{T_n - \mathbb{E}T_n}{n} = \sum_{j=2}^{n} \frac{\xi_j}{j}, \tag{39}$$

and

$$\frac{\mathrm{Var}(T_n)}{n^2} = 2 - H_n^{(2)} - \frac{H_n}{n}, \qquad H_n^{(2)} := \sum_{k=1}^{n} \frac{1}{k^2}. \tag{40}$$

Consequently,

$$\frac{T_n - n(H_n - 1)}{n} \longrightarrow Z_0 \qquad \text{almost surely and in } L^2,$$

and

$$\mathrm{Var}(T_n) \sim \left(2 - \frac{\pi^2}{6}\right) n^2. \tag{41}$$

Proof. The expectation follows from Theorem 4.1:

$$\mathbb{E}T_n = \sum_{j=1}^{n-1} \frac{n-j}{j} = nH_{n-1} - (n-1) = n(H_n - 1).$$

Since $p_j = 1/j$, the coefficient in Theorem 5.2 simplifies to

$$1 + (n-j)p_j = 1 + \frac{n-j}{j} = \frac{n}{j},$$

which proves (39).

For the innovation variances, recurrence (29) becomes

$$v_{n+1} = \left(1 - \frac{1}{n^2}\right) v_n + \frac{1}{n}\left(1 - \frac{1}{n}\right), \qquad v_2 = 0.$$

A direct induction gives

$$v_n = \frac{n}{n-1} H_{n-1} - 2, \qquad n \geq 2. \tag{42}$$

By (39) and orthogonality,

$$\frac{\mathrm{Var}(T_n)}{n^2} = \sum_{j=2}^{n} \frac{v_j}{j^2}.$$

Applying identity (28) with $n+1$ yields

$$\sum_{j=2}^{n} \frac{v_j}{j^2} = \mathrm{Var}(D_{n+1}) - v_{n+1}.$$

Here Proposition 2.1 and (42) give

$$\mathrm{Var}(D_{n+1}) = H_n - H_n^{(2)}, \qquad v_{n+1} = \frac{n+1}{n} H_n - 2.$$

Subtraction proves (40). Finally, $H_n^{(2)} \to \pi^2/6$ and $H_n/n \to 0$, while the asserted almost-sure and L^2 convergence is the specialization of Corollary 6.3. □

The limiting result for the uniform recursive tree is classical (Mahmoud, 1991; Dobrow and Fill, 1999). Corollary 6.5 is included to show that the general innovation framework specializes exactly to this benchmark; no novelty is claimed for the classical limit itself.

Combining Theorem 4.1 and Corollary 6.3 gives the pathwise average-case complexity.

Corollary 6.6 (Pathwise phase transition)**.** *The following convergences hold almost surely and in L^2:*

$$\alpha > -1: \quad \frac{T_n}{n} - (\alpha+1)\log n \longrightarrow \kappa_\alpha + Z_\alpha,$$

$$\alpha = -1: \quad \frac{T_n}{n} - \log\log n \longrightarrow \kappa_{-1} + Z_{-1},$$

$$\alpha < -1: \quad \frac{T_n}{n} \longrightarrow \mu_\alpha + Z_\alpha.$$

In particular,

$$\frac{T_n}{n \log n} \to \alpha + 1 \quad \textit{a.s. for } \alpha > -1,$$

and

$$\frac{T_n}{n \log\log n} \to 1 \quad \textit{a.s. for } \alpha = -1.$$

Remark 6.7. *The limit theorem distinguishes two levels of behavior. The dominant complexity scale is deterministic: logarithmic, iterated logarithmic, or constant. After removing that scale, however, an order-one random correction survives in the average cost, equivalently an order-n random correction survives in total path length.*

7 The summable old-age phase

When $\alpha < -1$, the total attachment weight is finite:

$$W_\infty := \sum_{i=1}^{\infty} i^\alpha = \zeta(-\alpha).$$

This permits a more concrete interpretation of the limit in Corollary 6.6.

Theorem 7.1 (Random constant-time limit)**.** *Let $\alpha < -1$ and define*

$$A_\infty := \sum_{i=1}^{\infty} i^\alpha D_i.$$

Then $A_\infty < \infty$ almost surely and

$$\frac{T_n}{n} \longrightarrow 1 + \frac{A_\infty}{\zeta(-\alpha)} \tag{43}$$

almost surely and in L^2. Moreover,

$$Z_\alpha = \frac{A_\infty - \mathbb{E}A_\infty}{\zeta(-\alpha)}$$

and

$$\mu_\alpha = 1 + \frac{\mathbb{E}A_\infty}{\zeta(-\alpha)}.$$

Proof. For $\alpha < -1$, Proposition 2.1 and Lemma 2.3 imply $\sup_i \mathbb{E}D_i < \infty$. Hence

$$\mathbb{E}\left[\sum_{i=1}^{\infty} i^\alpha D_i\right] = \sum_{i=1}^{\infty} i^\alpha \mathbb{E}D_i < \infty.$$

Thus $A_\infty < \infty$ almost surely and $A_n \uparrow A_\infty$ in L^1.

Since $W_n \to \zeta(-\alpha)$,

$$M_n = \frac{A_n - \mathbb{E}A_n}{W_n} \longrightarrow \frac{A_\infty - \mathbb{E}A_\infty}{\zeta(-\alpha)}$$

in probability. But Corollary 6.3 gives $M_n \to Z_\alpha$ almost surely and in L^2, so the two limits agree.

Next, from (23),

$$\mathbb{E}D_{n+1} = 1 + \frac{\mathbb{E}A_n}{W_n}.$$

The left side tends to $\sum_{j\geq 1} p_j = \mu_\alpha$, while the right side tends to $1 + \mathbb{E}A_\infty/\zeta(-\alpha)$. Therefore

$$\mu_\alpha = 1 + \frac{\mathbb{E}A_\infty}{\zeta(-\alpha)}.$$

Substitution in Corollary 6.6 gives (43). The L^2 convergence follows from that corollary. □

Thus the old-age regime has constant average traversal complexity but is not self-averaging: early random attachment decisions remain visible in the limiting cost.

8 Exact second-order analysis for young-age preference

We return to $\alpha = 1$. In this case

$$p_j = \frac{2}{j+1}.$$

Lyon and Mahmoud (2020) showed that the insertion depth has the same distribution as the insertion depth in a random binary search tree and, in particular,

$$\mathrm{Var}(D_n) = 2(H_n + 1) - 4H_n^{(2)}, \tag{44}$$

where

$$H_n^{(2)} := \sum_{k=1}^{n} \frac{1}{k^2}.$$

The martingale innovations make it possible to determine the global path-length variance constant explicitly.

Proposition 8.1 (Exact innovation variance)**.** *For the young-age tree and every $n \geq 2$,*

$$v_n = \frac{2(n+1)(n+2)}{n(n-1)} \left(H_n - 3 + \frac{6}{n+2} \right). \tag{45}$$

Consequently,

$$v_n = 2H_n - 6 + o(1).$$

Proof. For $p_n = 2/(n+1)$, recurrence (29) becomes

$$v_{n+1} = \left(1 - \frac{4}{(n+1)^2}\right) v_n + \frac{2(n-1)}{(n+1)^2}, \qquad v_2 = 0.$$

Direct substitution shows that (45) satisfies this recurrence and the initial condition. The asymptotic statement follows from $H_n = \log n + \gamma + o(1)$. □

Theorem 8.2 (Young-age path-length variance)**.** *For every $n \geq 2$,*

$$\mathrm{Var}(T_n) = \sum_{j=2}^{n} \left(\frac{2n - j + 1}{j+1} \right)^2 v_j, \tag{46}$$

where v_j is given by (45)*. Furthermore,*

$$\mathrm{Var}(T_n) \sim \left(8 - \frac{2\pi^2}{3}\right) n^2. \tag{47}$$

Proof. Since $p_j = 2/(j+1)$,

$$1 + (n-j)p_j = 1 + \frac{2(n-j)}{j+1} = \frac{2n - j + 1}{j+1}.$$

Equation (46) is therefore the specialization of (27).

For the limit constant, identity (28) gives

$$\sum_{j=2}^{n-1} p_j^2 v_j = \mathrm{Var}(D_n) - v_n.$$

Using (44) and Proposition 8.1,

$$\mathrm{Var}(D_n) - v_n = (2H_n + 2 - 4H_n^{(2)}) - (2H_n - 6 + o(1)).$$

Since $H_n^{(2)} \to \pi^2/6$,

$$\sigma_1^2 = \sum_{j=2}^{\infty} p_j^2 v_j = 8 - \frac{2\pi^2}{3}.$$

Corollary 6.3 now gives (47). □

Numerically,

$$8 - \frac{2\pi^2}{3} = 1.4202637326\ldots.$$

8.1 Same mean as a random BST, different fluctuations

Let P_n^{BST} be the internal path length of a random binary search tree constructed by inserting n distinct keys in uniformly random order. Classical analysis gives

$$\mathbb{E}P_n^{\mathrm{BST}} = 2(n+1)H_n - 4n, \tag{48}$$

$$\mathrm{Var}(P_n^{\mathrm{BST}}) \sim \left(7 - \frac{2\pi^2}{3}\right) n^2. \tag{49}$$

See, for example, Knuth (1998) and Flajolet and Sedgewick (2009).

Combining (21), (47), (48), and (49) yields the following.

Corollary 8.3 (Local equality, global separation)**.** *For every n,*

$$\mathbb{E}T_n = \mathbb{E}P_n^{\mathrm{BST}},$$

but

$$\frac{\mathrm{Var}(T_n)}{n^2} - \frac{\mathrm{Var}(P_n^{\mathrm{BST}})}{n^2} \longrightarrow 1.$$

Thus the young-age tree and random BST have identical expected total path length but distinct linear-scale fluctuations.

	Young-age tree	Random BST
Insertion-depth law	same as BST	same as young-age
$\mathbb{E}$ total path length	$2(n+1)H_n - 4n$	$2(n+1)H_n - 4n$
Asymptotic variance coefficient	$8 - \frac{2\pi^2}{3}$	$7 - \frac{2\pi^2}{3}$

The equality of insertion-depth marginals therefore hides a difference in the dependence structure across insertion times. The total path length, being a sum of strongly dependent depths, exposes this difference at second order.

9 Root-traversal costs

The results can be stated directly in terms of a simple online parent-pointer data structure. Each insertion stores one parent pointer. Consider the operation

```
TRACE-TO-ROOT(v)
  cost ← 0
  while v ≠ root do
    v ← parent[v]
    cost ← cost + 1
  return cost
```

For a uniformly sampled stored vertex U_n, the conditional expected number of pointer traversals is T_n/n by (13). Hence Corollary 6.6 gives not only an annealed expected complexity but a pathwise statement about a typical realized data structure.

Corollary 9.1. *For a realized power-weight recursive tree,*

$$\mathbb{E}(D_{U_n} \mid T_n^{(\alpha)}) = \begin{cases} (\alpha+1)\log n + \kappa_\alpha + Z_\alpha + o(1), & \alpha > -1, \\ \log\log n + \kappa_{-1} + Z_{-1} + o(1), & \alpha = -1, \\ \mu_\alpha + Z_\alpha + o(1), & \alpha < -1, \end{cases}$$

almost surely.

There are two complementary messages. First, the deterministic leading complexity changes sharply at $\alpha = -1$. Second, the random tree-to-tree variation in the average query cost remains of constant order after the leading term is removed. Equivalently, total path-length fluctuations remain of order n in every regime.

The young-age case adds a generation result: by Corollary 3.2, a tree with the same nonuniform parent law can be sampled in linear total time without maintaining a table of cumulative weights. Thus both construction and subsequent root-traversal costs admit direct average-case analysis.

10 Discussion and open problems

The innovation identities apply to every positive deterministic weight sequence. The condition $p_n = O(n^{-1})$ is sufficient for a nondegenerate linear-scale limit and includes power weights on both sides of their summability threshold. For $\alpha > -1$, the almost-sure limit can also be recovered from existing weighted-profile theory. The present argument gives a direct common treatment that additionally includes the critical, summable, and irregular cumulative-weight cases, and yields explicit variance identities.

The coefficient $1 + (n-j)p_j$ describes how the innovation introduced at time j contributes to total path length at time n. Its form explains why fluctuations of order n persist even when the mean depth remains bounded. For linear weights, evaluating these identities gives a global distinction from random binary search trees despite equality of the insertion-depth marginals.

Several questions remain open for this approach.

(i) The condition $p_n = O(n^{-1})$ is sufficient, but need not be necessary. It would be useful to characterize more general deterministic weights that admit a nondegenerate limit on the same scale.

(ii) The distribution of Z_α is represented here by a martingale series. Much more is known in the uniform case (Mahmoud, 1991; Dobrow and Fill, 1999); fixed-point descriptions or higher-moment recurrences for general power weights would complement the present analysis.

(iii) The regularity, monotonicity, and asymptotic behavior of $\alpha \mapsto \sigma_\alpha^2$ deserve separate study.

(iv) Other global functionals, such as the Wiener index or profile width, may further distinguish linear-weight recursive trees from random binary search trees.

11 Conclusion

We gave an innovation-based analysis of total path length for deterministic weighted recursive trees. The exact representation and variance recurrence lead to almost-sure and L^2 convergence

of centered total path length divided by n whenever $p_n = O(n^{-1})$. The limiting variance is finite and strictly positive. This provides a common elementary treatment without a polynomial cumulative-weight asymptotic.

For power weights, the theorem applies to every real exponent. It complements the existing profile-based limit theory for $\alpha > -1$ and covers the critical and summable regimes $\alpha \leq -1$. In the summable regime the limiting average depth is expressed through the weighted depths of the infinite tree. For linear weights the variance coefficient is $8 - 2\pi^2/3$, exceeding that of a random binary search tree by exactly one although all insertion-depth marginals agree. The same results describe random corrections to the average cost of tracing parent pointers to the root.

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